Answers to the questions

1)Connectivity

a)

```
Term2
/ # traceroute -n -q 1 192.180.30.3
traceroute to 192.180.30.3 (192.180.30.3), 30 hops max, 46 byte packets
 1  192.180.30.3  1.058 ms
/ # traceroute 192.180.40.3
traceroute to 192.180.40.3 (192.180.40.3), 30 hops max, 46 byte packets
 1  192.180.40.3 (192.180.40.3)  1.452 ms  0.771 ms  0.744 ms
/ # traceroute 170.2.0.3
traceroute to 170.2.0.3 (170.2.0.3), 30 hops max, 46 byte packets
 1  192.180.40.2 (192.180.40.2)  67.255 ms  23.209 ms  9.970 ms
 2  195.70.40.25 (195.70.40.25)  42.953 ms  17.112 ms  20.608 ms
 3  170.2.0.3 (170.2.0.3)  60.085 ms  14.742 ms  10.416 ms
/ #
```

For the traceroute from Term2 to RLin, the destinations 192.180.30.3 and 192.180.40.3 are reached in a single hop because they belong to networks directly connected to Term2. In contrast, the destination 170.2.0.3 requires three hops because it is not directly connected, so the traffic follows the default route via RCis2, then is forwarded to RCis1, which is directly connected to the 170.2.0.0/16 network, and finally reaches RLin.

b)

i

```
Term2    RCis2    Term1
/ # traceroute -n -q 1 192.180.30.44
traceroute to 192.180.30.44 (192.180.30.44), 30 hops max, 46 byte packets
 1  170.2.0.3  0.606 ms
 2  *
 3  *
 4  *
 5^C
/ # traceroute -n -q 1 192.180.40.44
traceroute to 192.180.40.44 (192.180.40.44), 30 hops max, 46 byte packets
 1  170.2.0.1  22.296 ms
 2  *
 3  *
 4  *
 5  *
 6^X  *
 7  *
 8  *
 9  *
10^Z[1]+  Stopped                 traceroute -n -q 1 192.180.40.44
/ # ^C
```

Without a route to 170.2.0.0/16 on RCis2, the forward path may still exist, but the return path from Term2 to Term1 fails. For 192.180.30.44, Term1 sends traffic through its default route via RLin. For 192.180.40.44, Term1 uses the specific route via RCis1. In both cases, Term2 replies through its default

gateway RCis2, which cannot reach 170.2.0.0/16, so the traceroute does not receive the return ICMP messages

ii.

RCis2(config)#ip route 170.2.0.0 255.255.0.0 195.70.40.25

```
Term2        RCis2        Term1
/ # traceroute -n -q 1 192.180.30.44
traceroute to 192.180.30.44 (192.180.30.44), 30 hops max, 46 byte packets
 1  170.2.0.3  1.802 ms
 2  192.180.30.44  19.283 ms
/ # traceroute -n -q 1 192.180.40.44
traceroute to 192.180.40.44 (192.180.40.44), 30 hops max, 46 byte packets
 1  170.2.0.1  3.098 ms
 2  195.70.40.26  31.078 ms
 3  192.180.40.44  31.323 ms
/ #
```

After adding the static route to the 170.2.0.0/16 network on RCis2 via RCis1, the traceroute works correctly because full connectivity is restored. Previously, the forward path existed but the return path was missing, preventing replies from reaching the source. With the route configured, packets can now be properly forwarded through RCis1 to reach the 170.2.0.0/16 network, allowing the traceroute to complete successfully.

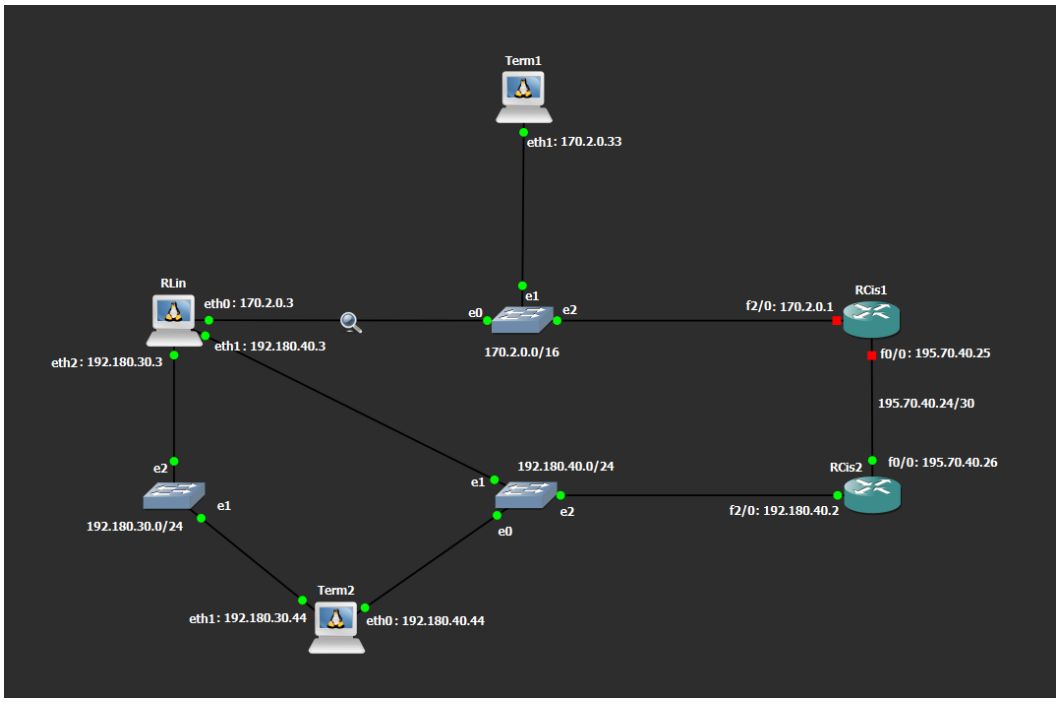
iii.

RCis2(config)#ip route 170.2.0.0 255.255.0.0 192.180.40.3

```
Term2        RCis2        Term1
/ # traceroute -n -q 1 192.180.30.44
traceroute to 192.180.30.44 (192.180.30.44), 30 hops max, 46 byte packets
 1  170.2.0.3  2.075 ms
 2  192.180.30.44  10.241 ms
/ # traceroute -n -q 1 192.180.40.44
traceroute to 192.180.40.44 (192.180.40.44), 30 hops max, 46 byte packets
 1  170.2.0.1  10.411 ms
 2  *
 3  192.180.40.44  27.762 ms
/ #
```

With the route to 170.2.0.0/16 on RCis2 changed to use RLin, the traceroute to 192.180.30.44 still shows two hops because the forward path from Term1 remains unchanged: the packet follows the default route to RLin, which is directly connected to the 192.180.30.0/24 network and can reach the destination in the next hop. The traceroute to 192.180.40.44 still requires three hops because Term1 uses the specific route to 192.180.40.0/24 via RCis1, so the packet still passes through RCis1 and RCis2 before reaching Term2. However, the second hop does not appear because RCis2 does not return the expected ICMP reply to the source in this routing condition, even though it is still part of the path.

c.



d.

When RCis1 is turned off, the behavior of the traceroute depends on the path used to reach each destination. For 192.180.30.44, the traceroute still succeeds because Term1 uses its default route via RLin, which is directly connected to the 192.180.30.0/24 network, so the path does not depend on RCis1. In contrast, the traceroute to 192.180.40.44 fails because Term1 uses a specific route via RCis1 to reach the 192.180.40.0/24 network. Since RCis1 is no longer available, the next hop becomes unreachable and the packet cannot be forwarded, resulting in a “host unreachable” condition.

e.

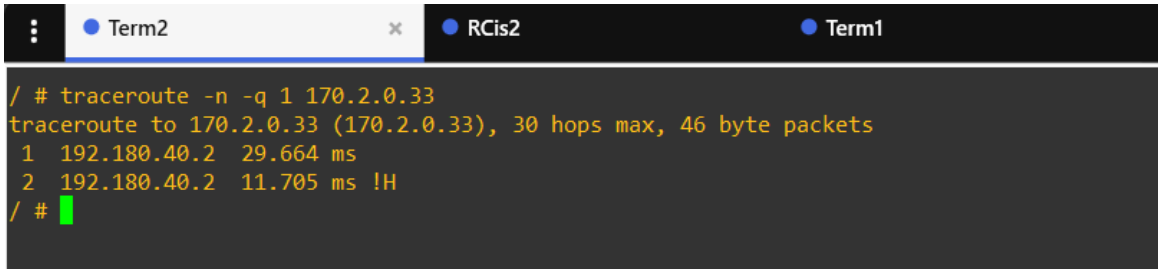
In this lab, dynamic routing would be advantageous because there are alternative paths between some networks, but static routing forces traffic to follow manually defined routes even when those routes become unavailable or inefficient. For example, after RCis1 was turned off, Term1 still tried to reach the 192.180.40.0/24 network through the specific static route via RCis1, so connectivity was lost even though another path through RLin still existed. With dynamic routing, the routing tables could be

updated automatically and traffic could be redirected through RLin instead of continuing to use the failed path. Likewise, traffic from 192.180.30.44 to Term1 could potentially use a shorter or more appropriate path depending on the available routes, instead of always following the manually configured default path through RCis2. This shows that dynamic routing is useful when a network has multiple possible paths and may suffer topology changes or failures

f.

i.

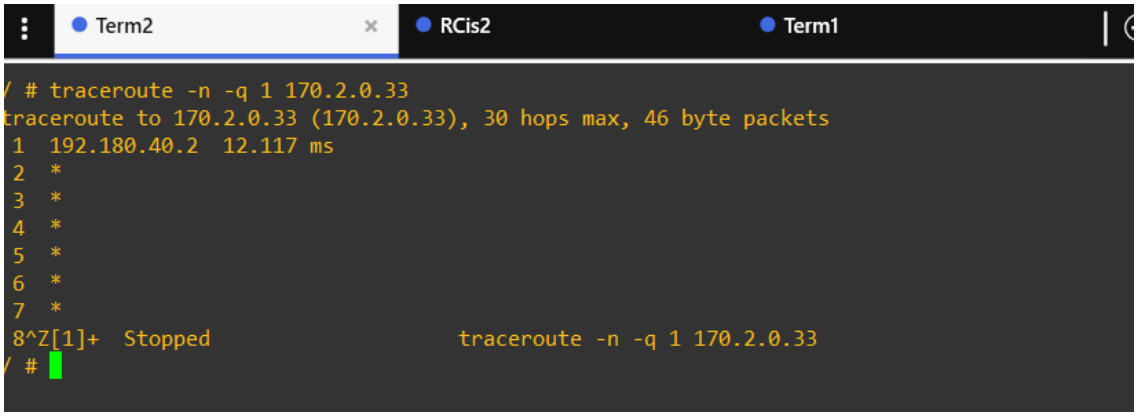
RCis2(config)#no ip route 170.2.0.0 255.255.0.0 192.180.40.3



In this scenario, Term2 sends the packet to its default gateway (RCis2). Since RCis2 does not have a route to the 170.2.0.0/16 network, it cannot forward the packet and responds with an ICMP "Host Unreachable" message. This is why the traceroute stops at RCis2 and shows the !H indication.

ii.

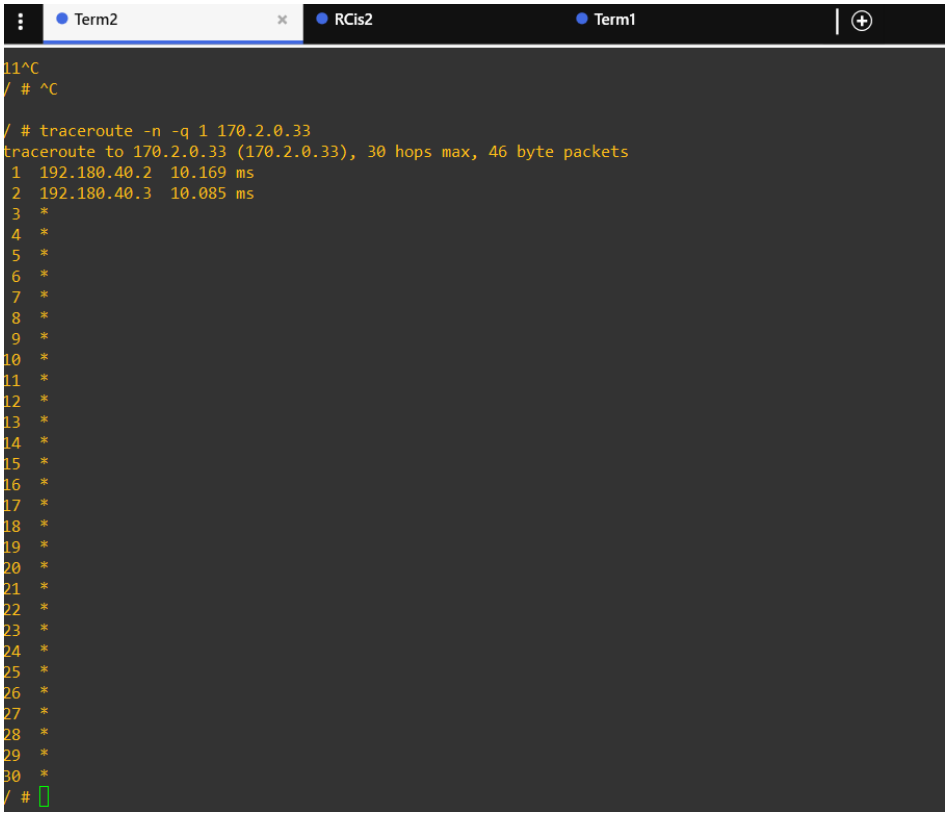
RCis2(config)#ip route 170.2.0.0 255.255.0.0 195.70.40.25



In this scenario, Term2 sends the packet to RCis2, so the first hop is still visible. However, although RCis2 has a static route to 170.2.0.0/16 via RCis1, RCis1 is turned off, so the next hop is no longer available. As a result, RCis2 cannot successfully forward the packet and no further replies are received, which is why the remaining hops appear as *.

iii.

RCis2(config)#ip route 170.2.0.0 255.255.0.0 192.180.40.3



In this scenario, Term2 sends the packet to RCis2, so the first hop corresponds to RCis2. Since RCis2 now has a static route to 170.2.0.0/16 via RLin, the packet is forwarded to RLin, which appears as the second hop. From that point onward, the packet reaches the destination network, but the return path is still dependent on RCis1, which is turned off. As a result, the replies cannot make their way back to the source, and no further responses are received, which is why all subsequent hops appear as *.

g.

Yes, dynamic routing would be useful in this scenario. With static routing, the routers continue to use predefined paths even when a link or device fails, such as RCis1 being turned off, which breaks the return path and prevents traceroute from receiving replies. If dynamic routing were used, the routers would detect that RCis1 is unavailable and automatically update their routing tables to use an alternative path, such as through RLin. This would restore end-to-end connectivity and allow traceroute to receive responses successfully.

h.

i.

Term2 | RCis2 | Term1

```
/ # traceroute -n -q 1 170.2.0.6
traceroute to 170.2.0.6 (170.2.0.6), 30 hops max, 46 byte packets
 1  192.180.40.2  5.301 ms
 2  192.180.40.3  13.097 ms
 3  192.180.40.3  3061.109 ms !H
/ # traceroute -n -q 1 170.2.0.6
traceroute to 170.2.0.6 (170.2.0.6), 30 hops max, 46 byte packets
 1  192.180.40.2  9.020 ms
 2  192.180.40.3  11.409 ms
 3  192.180.40.3  3072.486 ms !H
/ # traceroute -n -q 1 170.2.0.8
traceroute to 170.2.0.8 (170.2.0.8), 30 hops max, 46 byte packets
 1  192.180.40.2  8.559 ms
 2  192.180.40.3  9.452 ms
 3  192.180.40.3  3076.750 ms !H
/ # traceroute -n -q 1 170.2.0.8
traceroute to 170.2.0.8 (170.2.0.8), 30 hops max, 46 byte packets
 1  192.180.40.2  8.879 ms
 2  192.180.40.3  13.189 ms
 3  192.180.40.3  3060.266 ms !H
/ # ^C

/ # traceroute -n -q 1 170.2.0.50
traceroute to 170.2.0.50 (170.2.0.50), 30 hops max, 46 byte packets
 1  192.180.40.2  11.160 ms
 2  192.180.40.3  10.590 ms
 3  192.180.40.3  3061.432 ms !H
/ #
```

Capturing from Standard input [192.180.40.0/24 Ethernet0 to Term2 eth0]

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

icmp

No.	Time	Source	Destination	Protocol	Length	Info
4	56.804556	192.180.40.2	192.180.40.44	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
6	56.817481	192.180.40.2	192.180.40.44	ICMP	70	Redirect (Redirect for network)
7	56.817872	192.180.40.3	192.180.40.44	ICMP	88	Time-to-live exceeded (Time to live exceeded in transit)
9	59.879051	192.180.40.3	192.180.40.44	ICMP	88	Destination unreachable (Host unreachable)
15	97.627440	192.180.40.2	192.180.40.44	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
17	97.638911	192.180.40.2	192.180.40.44	ICMP	70	Redirect (Redirect for network)
18	97.639317	192.180.40.3	192.180.40.44	ICMP	88	Time-to-live exceeded (Time to live exceeded in transit)
20	100.711605	192.180.40.3	192.180.40.44	ICMP	88	Destination unreachable (Host unreachable)
27	123.991947	192.180.40.2	192.180.40.44	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
29	124.002006	192.180.40.2	192.180.40.44	ICMP	70	Redirect (Redirect for network)
30	124.002424	192.180.40.3	192.180.40.44	ICMP	88	Time-to-live exceeded (Time to live exceeded in transit)
32	127.079140	192.180.40.3	192.180.40.44	ICMP	88	Destination unreachable (Host unreachable)
34	133.893152	192.180.40.2	192.180.40.44	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
36	133.905380	192.180.40.2	192.180.40.44	ICMP	70	Redirect (Redirect for network)
37	133.905528	192.180.40.3	192.180.40.44	ICMP	88	Time-to-live exceeded (Time to live exceeded in transit)
39	136.967147	192.180.40.3	192.180.40.44	ICMP	88	Destination unreachable (Host unreachable)
54	589.896589	192.180.40.2	192.180.40.44	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
56	589.907217	192.180.40.2	192.180.40.44	ICMP	70	Redirect (Redirect for network)
57	589.907375	192.180.40.3	192.180.40.44	ICMP	88	Time-to-live exceeded (Time to live exceeded in transit)
59	592.968618	192.180.40.3	192.180.40.44	ICMP	88	Destination unreachable (Host unreachable)

```
▸ Frame 4: Packet, 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface -, id 0
▸ Ethernet II, Src: ca:02:13:65:00:38 (ca:02:13:65:00:38), Dst: 02:42:b4:b1:b1:00 (02:42:b4:b1:b1:
▸ Internet Protocol Version 4, Src: 192.180.40.2, Dst: 192.180.40.44
▾ Internet Control Message Protocol
  ▸ Type: Time-to-live exceeded (11)
    Code: 0 (Time to live exceeded in transit)
    Checksum: 0x8915 [correct]
    [Checksum Status: Good]
    Unused: 00000000
  ▾ Internet Protocol Version 4, Src: 192.180.40.44, Dst: 170.2.0.6
      0100 .... = Version: 4
      .... 0101 = Header Length: 20 bytes (5)
    ▸ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
      Total Length: 46
      Identification: 0x4b83 (19331)
    ▸ 010. .... = Flags: 0x2, Don't fragment
      ...0 0000 0000 0000 = Fragment Offset: 0
    ▸ Time to Live: 1
      Protocol: UDP (17)
      Header Checksum: 0x9b53 [validation disabled]
      [Header checksum status: Unverified]
      Source Address: 192.180.40.44
      Destination Address: 170.2.0.6
      [Stream index: 0]
  ▸ User Datagram Protocol, Src Port: 49099, Dst Port: 33435

0000  02 42 b4 b1 b1 00 ca 02  13 65 00 38 08 00 45 c0
0010  00 38 00 32 00 00 ff 01  e9 3b c0 b4 28 02 c0 b4
0020  28 2c 0b 00 89 15 00 00  00 00 45 00 00 2e 4b 83
0030  40 00 01 11 9b 53 c0 b4  28 2c aa 02 00 06 bf cb
0040  82 9b 00 1a 29 69
```

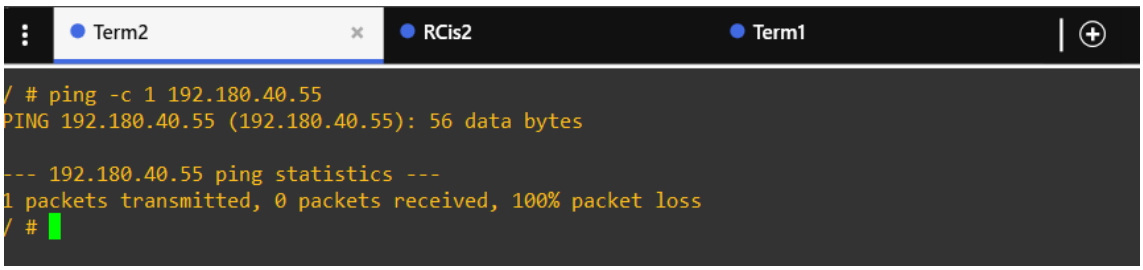
The output did not change because, although RCis2 sent ICMP Redirect messages indicating that RLin was a better next hop to reach the 170.2.0.0/16 network, Term2 did not start using that redirect information. As a result, both traceroute attempts still followed the same path: first RCis2, then RLin, and finally RLin returned “Host Unreachable” because the destination IP did not exist on the directly connected 170.2.0.0/16 network.

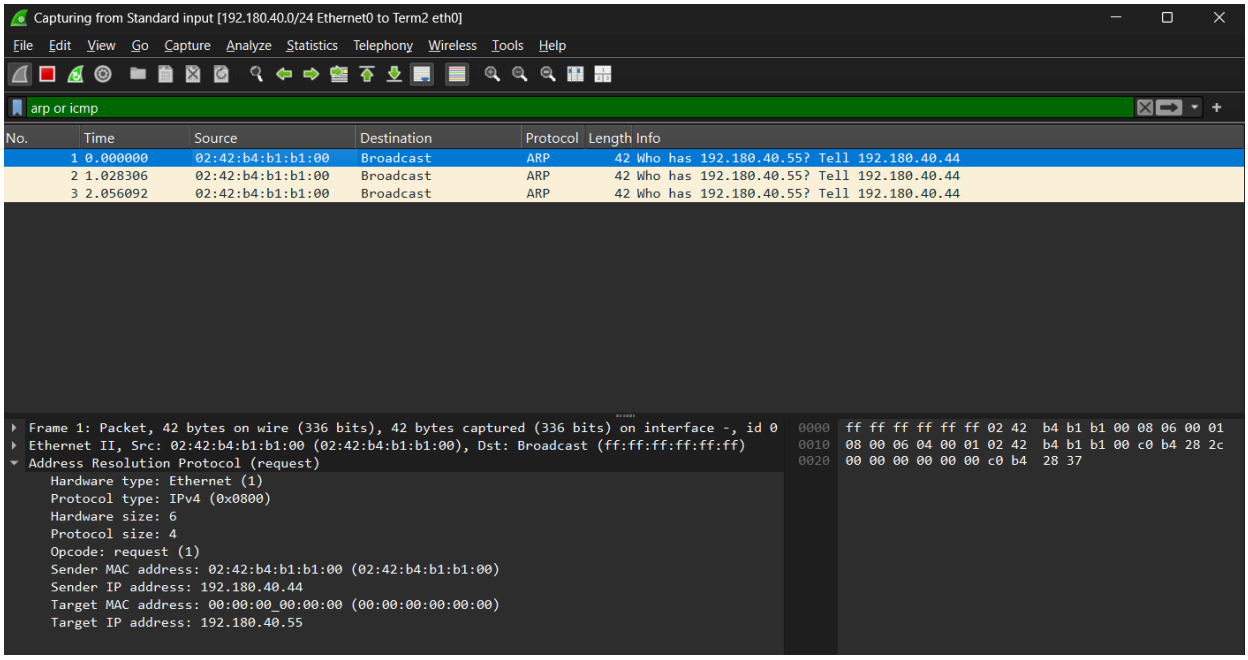
ii.

The output did not change because, although RCis2 sent ICMP Redirect messages indicating that RLin was a better next hop to reach the 170.2.0.0/16 network, Term2 did not start using that redirect information. As a result, both traceroute attempts still followed the same path: first RCis2, then RLin, and finally RLin returned “Host Unreachable” because the destination IP did not exist on the directly connected 170.2.0.0/16 network.

2. ARP

a.

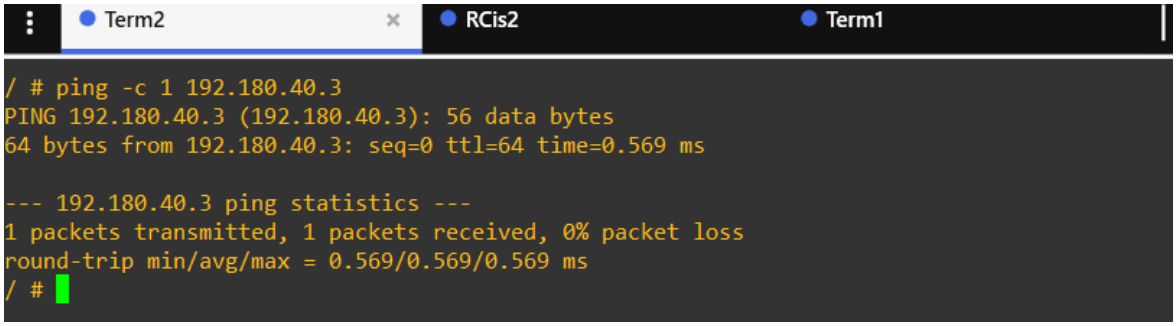


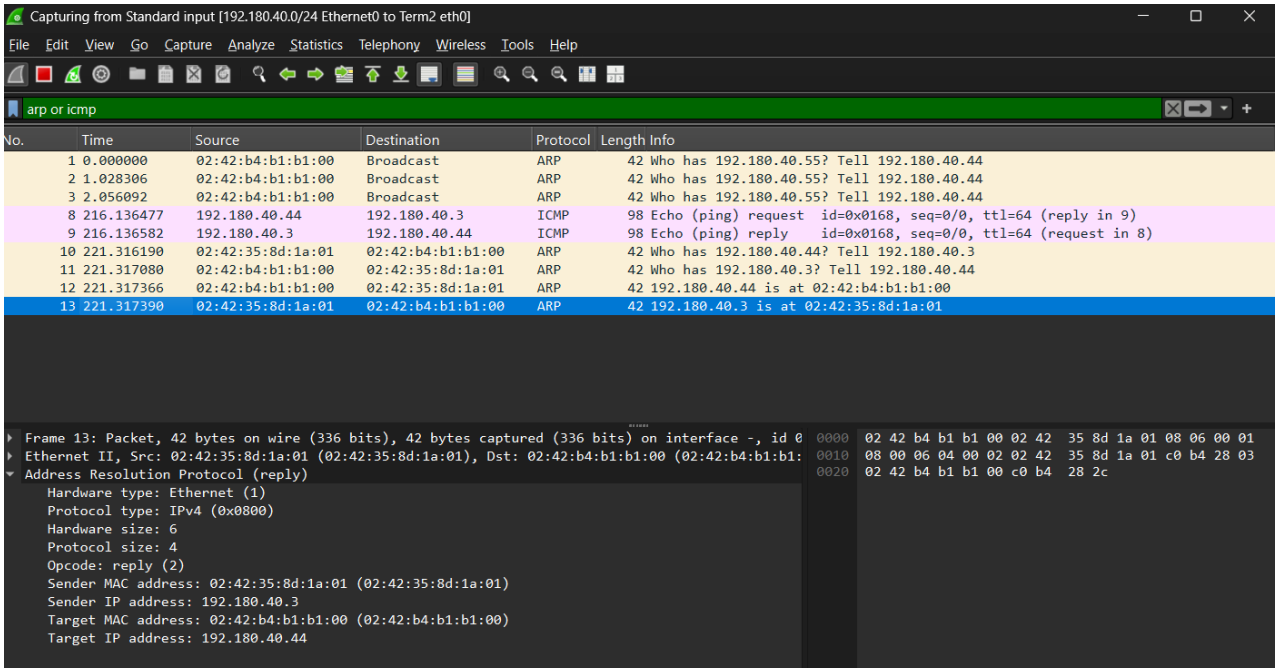


The capture shows only ARP requests because the destination IP 192.180.40.55 does not exist in the local network. Since Term2 is trying to reach a host in the same subnet, it must first resolve the

destination MAC address through ARP. As no device replies, ARP resolution fails, no ICMP Echo Request is sent, and the ping ends with 100% packet loss.

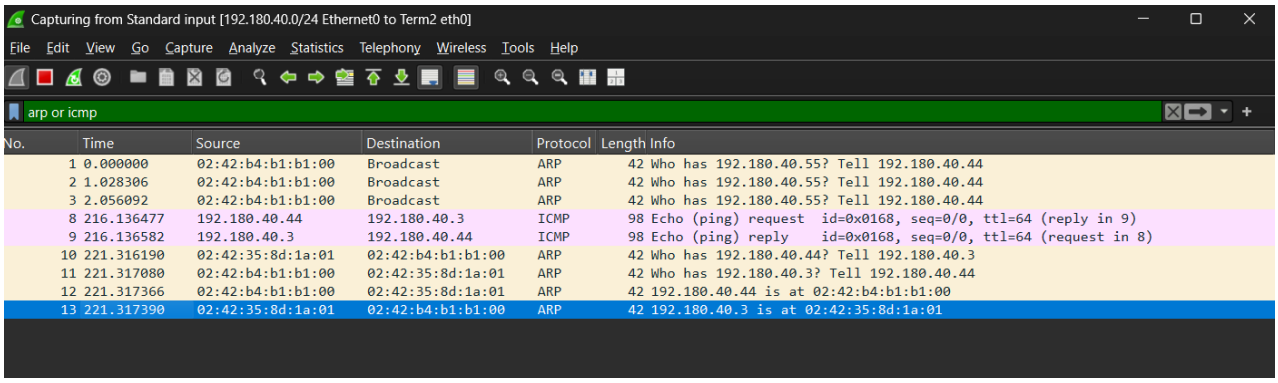
b.





In this case, the destination IP exists in the local network, so ARP resolution succeeds. Term2 first sends ARP requests to discover the MAC address of 192.180.40.3, and the target replies with its MAC address. After that, the ICMP Echo Request is sent and the Echo Reply is received successfully, which is why the ping completes with 0% packet loss.

c.



No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	02:42:b4:b1:b1:00	Broadcast	ARP	42	Who has 192.180.40.55? Tell 192.180.40.44
2	1.028306	02:42:b4:b1:b1:00	Broadcast	ARP	42	Who has 192.180.40.55? Tell 192.180.40.44
3	2.056092	02:42:b4:b1:b1:00	Broadcast	ARP	42	Who has 192.180.40.55? Tell 192.180.40.44
8	216.136477	192.180.40.44	192.180.40.3	ICMP	98	Echo (ping) request id=0x0168, seq=0/0, ttl=64 (reply in 9)
9	216.136582	192.180.40.3	192.180.40.44	ICMP	98	Echo (ping) reply id=0x0168, seq=0/0, ttl=64 (request in 8)
10	221.316190	02:42:35:8d:1a:01	02:42:b4:b1:b1:00	ARP	42	Who has 192.180.40.44? Tell 192.180.40.3
11	221.317080	02:42:b4:b1:b1:00	02:42:35:8d:1a:01	ARP	42	Who has 192.180.40.3? Tell 192.180.40.44
12	221.317366	02:42:b4:b1:b1:00	02:42:35:8d:1a:01	ARP	42	192.180.40.44 is at 02:42:b4:b1:b1:00
13	221.317390	02:42:35:8d:1a:01	02:42:b4:b1:b1:00	ARP	42	192.180.40.3 is at 02:42:35:8d:1a:01

The captures illustrate how ARP operates as a prerequisite for IP communication within the same subnet. When Term2 attempts to send traffic to a local IP address, it must first resolve the corresponding MAC address. This is done by broadcasting an ARP request asking “Who has this IP?”. If the destination host exists, it replies with its MAC address, allowing the sender to populate its ARP cache and proceed with normal communication using ICMP, as observed with the successful ping to 192.180.40.3, where ARP resolution is followed by an ICMP Echo Request and Echo Reply.

In contrast, when the destination IP does not exist (e.g., 192.180.40.55), ARP resolution fails. The sender transmits multiple ARP requests (retransmissions) because no reply is received. These retransmissions occur at defined intervals and are limited in number. After a timeout period with no response, the system concludes that the host is unreachable and does not send any ICMP packets, which explains why only ARP requests are observed and the ping results in 100% packet loss.

Additionally, ARP entries are stored temporarily in the ARP cache with a timeout. If an entry expires or is not yet present, ARP resolution is triggered again, which explains the additional ARP packets seen even after successful communication. This mechanism ensures that MAC address mappings remain updated while avoiding unnecessary ARP traffic.